

Hands-on 2 Classification

(Keras3/Tensorflow environnement)



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Preliminary remarks

- All four hands-on can be done on a computing cloud such as that of *CalculQuebec*, *Google Colab*, ... or on your own computer
- **For beginners** : we recommend you to start with *CalculQuebec*
- For advanced users with CPU/GPU access
 - You can run on your specific hardware
 - Tips for installing the required environnement are [here](#).

First, setup your *CalculQuebec* Account

(cf. DLMI 2026 hands on setup instructions)

1. *Create an account on CalculQuebec Cloud and log-in*
<https://jupyter.dmlti.calculquebec.cloud/hub/login>
2. Create a compute node with
 1. Memory at 4096MB
 2. GPU 1x1G.5GB
3. Start the jupyter-lab server and wait for it **Steps covered here**
4. Download the hands-on material into your jupyter-lab
5. Open a notebook
6. Cells philosophy and execution

Jupyter Lab interface (can change)

The screenshot shows the Jupyter Lab web interface. On the left is a file browser sidebar with a table of files and folders. Two orange arrows point to the 'upload' button (a square with an upward arrow) and the file list area. A red text box at the bottom left says 'Drag and drop files here (or via the upload button)'. The main area is titled 'Launcher' and contains several sections: 'Notebook' with icons for Python 3.11, LibreQDA, Mate Desktop (VNC), MLflow, OpenRefine, Python bundle, RStudio, Tensorboard, and VS Code; 'Console' with Python 3.11 and Python bundle; 'Other' with Terminal, Job statistics, Text File, Markdown File, Python File, and Show Contextual Help; and 'Documentation' with General and Jupyter. A green arrow points to the 'Launcher' tab at the top. Another green arrow points to the 'Terminal' icon in the 'Other' section. A small notification box in the bottom right corner says 'Would you like Jupyter news?'.

File Edit View Run Kernel Tabs Settings Help

← → ↻ 🔍 jupyter.dmlci.calculquebec.cloud/user/pmjodoin/lab

Launcher

Notebook

Python 3.11 LibreQDA [?] Mate Desktop (VNC) [?] MLflow [?] OpenRefine [?] Python bundle RStudio [?] Tensorboard [?] VS Code [?]

XFCE4 Desktop (VNC) [?]

Console

Python 3.11 Python bundle

Other

Terminal Job statistics Text File Markdown File Python File Show Contextual Help

Documentation

General Jupyter

Would you like Jupyter news?

Drag and drop files here (or via the upload button)

Launcher tab

Terminal

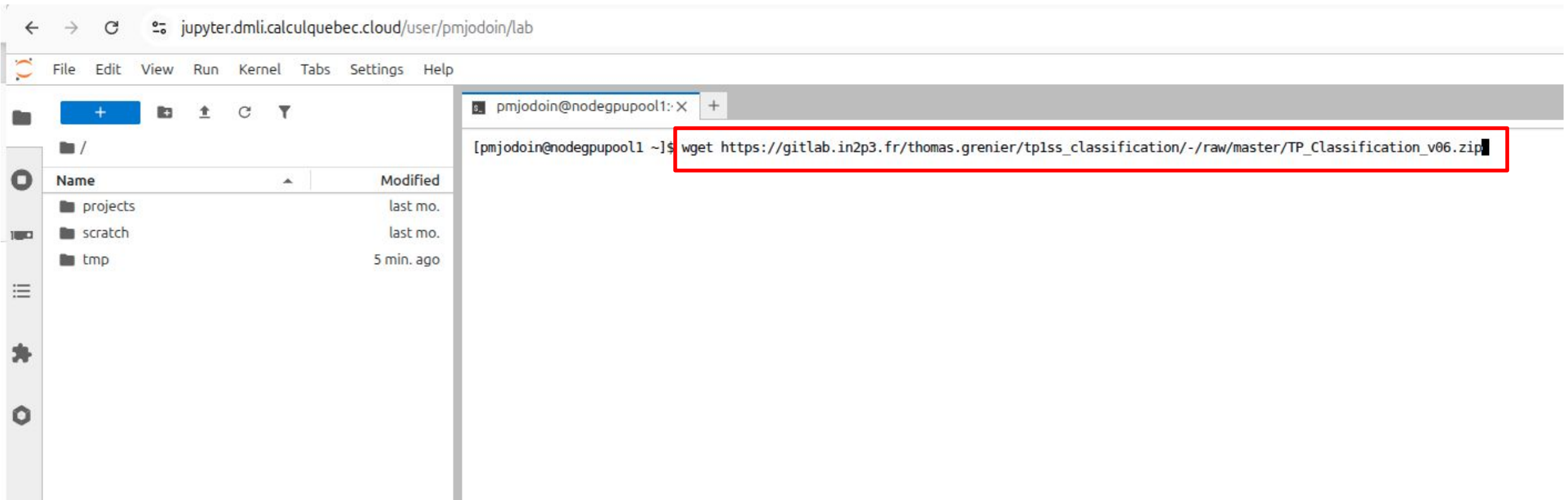
Download Hands-on materials

(same as for hands on 1)

a- Open a **Terminal** (c.f. previous page)

b- download from the terminal with the following command :

```
wget https://gitlab.in2p3.fr/thomas.grenier/tp1ss_classification/-/raw/master/TP_Classification_v06.zip
```



Download Hands-on materials

(same as for hands on 1)

a- Open a **Terminal** (c.f. previous page)

b- download from the terminal with the following command :

```
wget https://gitlab.in2p3.fr/thomas.grenier/tp1ss_classification/-/raw/master/TP_Classification_v06.zip
```

c- unzip from jupyter lab (in a Terminal)

```
unzip TP_Classification_v06.zip
```

The screenshot displays the JupyterLab interface. On the left, the file explorer shows a directory structure with files like '01_LoadData_K3.ipynb', '02a_ClassificationRF_K3.ipynb', '02b_ClassificationNN_K3.ipynb', '03_Interpretability_K3.ipynb', '04_LearningCurves_K3.ipynb', '05_HyperparameterOptimization...', '06_KnowledgeDistillation_K3.ipynb', '06_KnowledgeDistillationNoClass...', '09_GradAndAutoGrad.ipynb', 'toolsK3.py', and 'TP_Classification_v06.zip'. A red bracket on the left side of the file explorer points to the files from '01_LoadData_K3.ipynb' to '06_KnowledgeDistillationNoClass...', with the text 'unzipped files' next to it. The terminal on the right shows the execution of the 'wget' command to download 'TP_Classification_v06.zip' and the subsequent 'unzip TP_Classification_v06.zip' command, which is highlighted with a red box. The terminal output shows the files being inflated from the archive.

```
[pmjodoin@nodegypool1 ~]$ history
1 history
[pmjodoin@nodegypool1 ~]$ wget https://gitlab.in2p3.fr/thomas.grenier/tp1ss_classification/-/raw/master/TP_Classification_v06.zip
--2026-06-09 18:04:37-- https://gitlab.in2p3.fr/thomas.grenier/tp1ss_classification/-/raw/master/TP_Classification_v06.zip
Resolving gitlab.in2p3.fr... 134.158.69.41
Connecting to gitlab.in2p3.fr|134.158.69.41|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1129440 (1.1M) [application/octet-stream]
Saving to: 'TP_Classification_v06.zip'

TP_Classification_v06.zip      100%[=====] 1129440/1129440

2026-06-09 18:04:38 (2.02 MB/s) - 'TP_Classification_v06.zip' saved [1129440/1129440]

[pmjodoin@nodegypool1 ~]$ unzip TP_Classification_v06.zip
Archive: TP_Classification_v06.zip
  inflating: 01_LoadData_K3.ipynb
  inflating: 03_Interpretability_K3.ipynb
  inflating: 04_LearningCurves_K3.ipynb
  inflating: 05_HyperparameterOptimization.ipynb
  inflating: 06_KnowledgeDistillation_K3.ipynb
  inflating: 06_KnowledgeDistillationNoClass_K3.ipynb
  inflating: 09_GradAndAutoGrad.ipynb
  inflating: toolsK3.py
  inflating: figures/architecture_cnn_en.png
  inflating: figures/architecture_fcnn_en.png
  inflating: figures/architecture_resnet_en.png
  inflating: figures/decision_tree.png
  inflating: figures/Figure1_SagittalAxialCoronal_small.png
  inflating: figures/ResNet_TransferLearning.png
  inflating: figures/TensorBoard.png
  inflating: figures/treeL2.png
  inflating: 02a_ClassificationRF_K3.ipynb
  inflating: 02b_ClassificationNN_K3.ipynb
  inflating: 05_HyperparameterOptimization_K3.ipynb
[pmjodoin@nodegypool1 ~]$
```

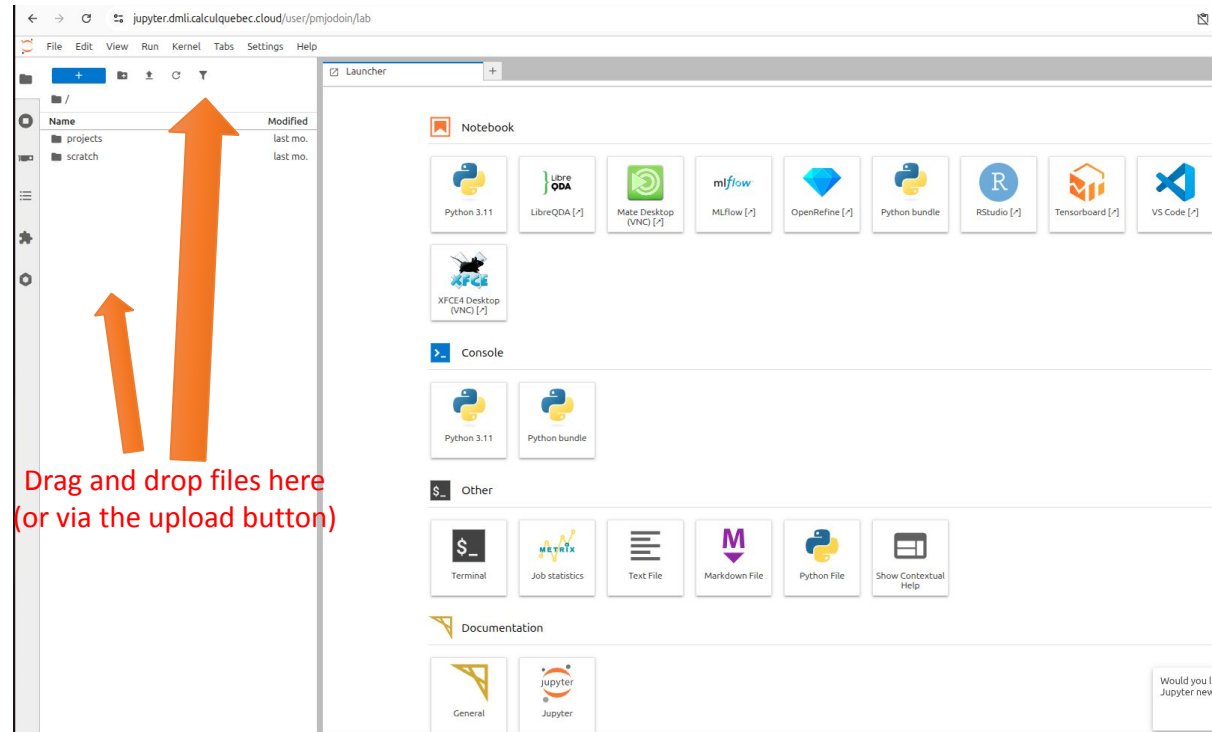
Download Hands-on materials

If the previous step did not work, you can

a) download the zip file on your computer

https://gitlab.in2p3.fr/thomas.grenier/tp1ss_classification/-/raw/master/TP_Classification_v06.zip

b) Upload it in your online environment



Download Hands-on materials

If the previous step did not work, you can

a) Download the zip file on your computer

https://gitlab.in2p3.fr/thomas.grenier/tp1ss_classification/-/raw/master/TP_Classification_v06.zip

b) Upload it in your online environment

c) Unzip the file

Hands on 2

For Hands on 2, please focus on the 7 last Notebooks

Name	Modified
data	39 sec. ago
figures	34 min. ago
projects	last mo.
scratch	last mo.
01_LoadData_K3.ipynb	now
02a_ClassificationRF_K3.ipynb	now
02b_ClassificationNN_K3.ipynb	now
03_Interpretability_K3.ipynb	now
04_LearningCurves_K3.ipynb	now
05_HyperparameterOptimizatio...	9 hr. ago
05_HyperparameterOptimizatio...	9 hr. ago
06_KnowledgeDistillation_K3.ipy...	8 hr. ago
06_KnowledgeDistillationNoClas...	9 hr. ago
09_GradAndAutoGrad.ipynb	2 days ago
toolsK3.py	yesterday
TP_Classification_v06.zip	18 min. ago

1. Ha

2. Int

Axial Sagittal Coronal

T1